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1 RESEARCH PROFILE

Applied Mathematician and Researcher specializing in the **mathematical analysis of Deep Learning algorithms**. My work focuses on **stability**, **robustness**, and **solvability** in high-dimensional spaces.

I have extensive experience in: 1. **Trustworthy AI:** Analyzing the robustness of neural image compression against perturbations and detecting hallucinations in LLMs via latent space geometry. 2. **Inverse Problems:** Using sparse-regularized optimization and PINNs to recover dynamical systems from noisy data. 3. **High-Performance Computing:** Implementing parallel algorithms (MPI/OpenMP) for large-scale numerical linear algebra.

I am eager to apply my background in **functional analysis** and **numerical optimization** to the Emmy Noether project on "Stability and Solvability in Deep Learning."

2 RESEARCH EXPERIENCE

Trustworthy AI: Hallucination Detection in LLMs 2025–2026
Skoltech, Russia

M.Sc. Thesis Project
Focus: Latent Space Geometry & Solvability.

- Investigated the **stability** of Qwen2.5 hidden states to distinguish truthful vs. hallucinated outputs.
- Developed `PeakLoc`, an attention-based method for localizing hallucinations in code generation, leveraging graph-theoretic properties of attention heads.
- Analyzed the **computability** of truthfulness by probing internal representations, linking geometric features in latent space to model reliability.

Robustness of Neural Image Compression 2024–2025
Skoltech, Russia

Presented at NeuroInformatics 2025
Focus: Stability under Non-linear Perturbations.

- Analyzed the **robustness** of autoencoder-based image compression pipelines against adversarial and non-linear noise.
- Proposed a Tanh-Arctanh perturbation framework to quantify stability margins, demonstrating how reconstruction error correlates with latent space topology.
- Evaluated downstream impact on computer vision tasks, providing metrics for **reliable** compressed data usage in scientific imaging.
- *Output:* Publication in NeuroInformatics 2025 Proceedings.

Sparse-Regularized System Identification 2025
Skoltech, Russia

Accepted at IEEE ADACIS'25
Focus: Inverse Problems & Regularization.

- Developed sparse-regularized models (Lasso/Elastic Net) to solve **inverse problems** in multiphase flow dynamics.
- Addressed **solvability** issues in under-determined systems by enforcing sparsity constraints on velocity fields.
- Implemented autoregressive delay-embedding to capture

1 EDUCATION

M.Sc. Advanced Computational Science *2024–2026*
Skoltech, Moscow, Russia

- **Thesis:** Trustworthy AI & Latent Space Geometry.
- **Focus:** High-dimensional Probability, Numerical Optimization, Deep Learning Theory.
- *Relevant Coursework:* Functional Analysis, Inverse Problems, TDA.

M.Sc. Mathematics *2022–2024*
Andhra University, India

- **Focus:** Linear Algebra, Real Analysis, Statistics.
- *Rank:* Top Student (1st Year).

B.Sc. Mathematics *2017–2021*
Univ. of Antananarivo, Madagascar

2 SELECTED PROJECTS

DefectZero: Unsupervised Anomaly Detection 2026
Personal Project

- Implemented **PaDiM** (Patch Distribution Modeling) for industrial defect detection.
- Modeled normal data distribution using Mahalanobis distance in EfficientNet feature space.
- Demonstrated **stability** of detection without requiring defective training samples (unsupervised).
- *Tech:* PyTorch, EfficientNet-B0, Statistical Modeling.

Parallel PCA for High-Dimensional Data 2025
Skoltech HPC

- Implemented parallelized PCA using **C++ with MPI/OpenMP**.
- Optimized SVD computation for large-scale matrices, reducing memory overhead and improving **computational efficiency**.

3 TECHNICAL SKILLS

Mathematics: High-Dimensional Probability, Functional Analysis, Sparse Regularization, SVD/Tensor Decompositions, Topological Data Analysis (TDA), Inverse Problems, Numerical Optimization.

Programming: Python (PyTorch, NumPy, SciPy), C/C++ (MPI/OpenMP for HPC), Bash, LaTeX, SQL.

Machine Learning: Trustworthy AI (Robustness/Hallucination), Neural Compression, Transformers, CNNs, Physics-Informed Neural Networks (PINNs), Self-Supervised Learning.

Tools & HPC: Linux/Unix, Git, Docker, Slurm, MPI/OpenMP Parallelization.

4 LANGUAGES

- **English**: Fluent (Professional Research)

spatio-temporal dependencies, balancing model complexity with generalization stability.
- *Output:* Comparative analysis of DMD vs. Sparse Regression for transient flow prediction.

Inverse Problems in ODEs via PINNs 2026
Skoltech, Russia

Under review at DAMDID 2026
Focus: Physics-Informed Machine Learning.
- Developed a regularized framework for recovering time-varying coefficients in dynamical systems from noisy observations.
- Addressed the **non-convexity** of PINN loss landscapes by introducing basis-function representations and automatic windowing.
- Enhanced **stability** of derivative estimation in high-noise regimes using weak formulation integral approaches.

3 WORK EXPERIENCE

Embedded ML Engineer Intern 2025
Embox, St Petersburg, Russia

- Optimized **TensorFlow Lite** and **OpenCV** for resource-constrained embedded systems.
- Focused on memory-efficient inference, ensuring **computational stability** of CV pipelines on edge devices.
- Gained practical experience in deploying mathematical models in real-world, low-latency environments.

Mathematics Teacher 2020–2021
CEG Betsimitatatra, Madagascar

- Taught advanced mathematics, developing the ability to communicate complex abstract concepts clearly.

- **French**: Fluent
- **Malagasy**: Native
- **German**: Beginner (Willing to learn)

5 CERTIFICATIONS

- Int. Youth Math Challenge Award (2022)
- Data Science for Engineers (IIT Madras, 2024)